

Chong Kang Rui

Software Engineer

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PROFESSIONAL SUMMARY

Full-stack developer skilled in JavaScript/TypeScript, React, Node.js/Express, Next.js and PostgreSQL, shipping production web apps end-to-end with REST APIs, layered architecture and relational data design. Brings 3 years of professional experience building performance-critical, real-time systems in Unreal Engine and C++, applying that debugging discipline and systems-level thinking to backend concurrency and architecture.

SKILLS

PROGRAMMING LANGUAGES:	JavaScript TypeScript Python C++ C#
MARKUP & STYLING:	HTML CSS Tailwind CSS Shadcn
FRAMEWORKS & LIBRARIES:	React Node.js Express.js Next.js FastAPI
TOOLS & TECHNOLOGIES:	Git RESTful APIs Github Action Docker
AI & RAG ENGINEERING:	RAG BM25 Cross-Encoder Reranking Vector Search
TESTING :	Jest Vitest Supertest Playwright React Testing Library
DATABASES:	MongoDB PostgreSQL
GAME DEVELOPMENT:	Unreal Engine Unity

Project

LegalRAGSummarizer - FastAPI, React, Groq, fastembed | [GitHub](#) | [Live Demo](#)

- Built a RAG system for querying legal contracts, implementing three retrieval strategies — vector search, BM25/vector hybrid via Reciprocal Rank Fusion, and cross-encoder reranking — with citation-grounded generation that validates every cited clause against retrieved context to catch fabricated citations before they reach the user.
- Designed an evaluation harness scoring precision@k, recall@k, MRR and citation accuracy against a hand-labeled golden set, exposing production bugs invisible to manual testing — including a tokenizer regex silently degrading BM25 to character-level noise and an unvalidated citation metric that scored hallucinated clause references as correct.
- Engineered an async streaming pipeline (SSE) over Groq's API with retry/backoff on rate-limit errors and CPU-bound reranking offloaded via `asyncio.to_thread`, keeping one client's in-flight generation from blocking other concurrent requests on the event loop.

EcommerGameShop - React, Node.js, Express, PostgreSQL | [GitHub](#) | [Live Demo](#)

- Built a full-stack e-commerce platform with Stripe checkout, JWT-based role authentication and an admin dashboard for managing products, orders, and refunds.

- Engineered concurrency-safe checkout logic, resolving race conditions in stock reservation and Stripe payment initialization via DB transactions, row-level locking and idempotent requests to prevent overselling and duplicate charges.
- Refactored a monolithic MVC backend into a layered Controller-Service-Repository architecture, decoupling business logic from HTTP and data-access concerns to improve testability and maintainability.
- Set up CI/CD and containerization. GitHub Actions running integration (Jest/Supertest) and unit (Vitest) tests on every PR, with Docker Compose for a reproducible one-command dev environment.

PROFESSIONAL EXPERIENCE

VR Software Engineer

Insense Studios - Remote | Sep, 2024 – May, 2026

- Designed the procedural level-generation system for a VR game - a constraint-based algorithm with backtracking that validates and places content automatically, the same class of problem behind scheduling and layout engines.
- Built the game's AI and interaction systems, modular, event-driven state machines driving NPC behavior, including internal tooling to tune and debug them at runtime.
- Engineered the core gameplay loop, synchronizing simultaneous state changes across combat, equipment, and movement in real time without race conditions or inconsistent behavior.
- Profiled and optimized rendering performance, improving frame rate by 140% (30 to 72 FPS) through systematic bottleneck analysis.

Software Engineer

Freelancer - Remote | May, 2024 – Aug, 2024

- Designed an extensible weapon and ability system architecture for a game project, letting new content be added without code duplication.
- Built a lifecycle management system to spawn, track, and retire in-game AI characters in a high-traffic real-time environment.

Software Engineer

Nimbus Games - Remote | Feb, 2023– Jan, 2024

- Delivered two commercial projects at Nimbus Games: Chiyo and a playable multiplayer demo supporting up to 4 concurrent players.
- Used Unreal Engine's replication system to keep gameplay and AI state consistent between server and clients for the multiplayer demo.
- Built the core gameplay systems for Chiyo - puzzle logic, AI behavior, the save system, and player interaction and inventory mechanics.

EDUCATION

Bachelor of Game Technology (Hons)

UOW KDU University College, Shah Alam, Selangor | 2019-2023

Cumulative GPA: 3.59, First Class Honours